Internet Topology RIS — Tool Quick Reference

*Collector range syntax: rrc00-rrc05,rrc11 · Dates: YYYYMMDD · **just**: variable assignments must come BEFORE the recipe name*

PIPELINE TOOLS

— uv run <tool> from repo root

get-mrt.py *Download RIPE RIS bview (00:00 UTC RIB snapshot)*

`--collectors rrc00-rrc25 --date 20240101 --output-dir ./mrt`

Skips already-downloaded files · failed collectors reported but do not abort

get-routeviews.py *Download RouteViews bview (.bz2)*

`--collectors route-views.amsix,ix-br2.gru,route-views2,route-views.sg`
`--date 20240101 --output-dir ./mrt`

download-ris-day.py *Download + extract bview AND all 288 five-minute updates per collector*

`--collectors rrc00-rrc25 --date 20240101`
`--mrt-dir ./mrt --output-dir ./paths --keep (keep update files) --concurrency 20`

Writes .full.gz.txt / .v4.txt / .v6.txt per collector — alternative to get-mrt + extract-paths

extract-paths.py *Extract AS paths from MRT files via bgpreader*

`--input-files "mrt/20240101/*.gz" --output-dir ./paths`

Per file: .gz.txt (all), .gz.v4.txt (IPv4), .gz.v6.txt (IPv6) · accepts .gz and .bz2 · skips complete sets

build-topology.py *Build deduplicated undirected edge list from AS path files*

`--input "paths/*.gz.txt" (repeatable) --output topology.txt`
`--filter-registered (drop unregistered ASNs via NRO stats) --date YYYYMMDD`
`--justify-output (zero-pad to 8 digits) --no-paths --workers N`

Outputs: topology.txt, -asns.txt, -paths.txt, [-dropped-asns.txt with --filter-registered]

topology-ops.py *Set operations on topology edge list files*

`--op add|sub --output out.txt FILE [FILE …] --justify-output`

add = union of all files · sub = first file minus all subsequent files

lookup-as.py *Query RIPEstat: holder name, prefix count, last seen*

`--asn N | --file asns.txt | stdin --output asn-info.tsv --concurrency 10`

fetch-results.sh *Rsync topology outputs from multiple remote machines*

Configure SOURCES (rsync URLs), INCLUDES (glob array), EXCLUDE at top of script
Output: ./fetch/<hostname>/ · continues on per-host failure · reports failed hosts

just recipes — `just date=20240101 collectors=rrc00-rrc25 pipeline`

`pipeline download download-routeviews ris-day full-day process topology topology-v4 topology-v6 pack-and-publish lab`

GRAPH ANALYSIS TOOLS

— uv run graph-analysis/<tool> from repo root

topology-to-gml.py *Convert topology edge list to NetworkX GML (Gephi, yEd, Cytoscape …)*

`--input topology.txt[.gz] --output topology.gml --min-degree N`
`--enrich (fetch AS holder names from RIPEstat) --enrich-min-degree 3 --concurrency 10`
`--cache PATH`

Node attrs: asn (int), label (holder name or ASN string) · name cache TTL 60 days at ~/.cache/internet-topology-ris/

gml-to-topology.py *Export GML back to two-column topology edge list (inverse of above)*

`--input topology.gml --output topology.txt --min-degree N --justify-output`

ASNs always read from node asn attribute — safe for enriched GML files where keys may be holder name strings

hubs.py *List ASNs with degree ≥ N (transit, IXPs, large content providers)*

`--input topology.txt[.gz] or .gml --min-degree N --output hubs.txt --wide`

plain: one ASN/line sorted numerically · --wide: two cols (ASN degree) sorted by degree desc · single-pass for .txt

topology-core.py *Extract degree-filtered core — edges where both endpoints have degree ≥ N*

`--input topology.txt[.gz] --min-degree N --output core.txt`
`--wide (add DEG1 DEG2 columns) --justify-output`

Two-pass, no GML needed · output is a valid topology file · use instead of topology-to-gml --output .txt

geo-filter.py *Filter to edges where both ASNs are assigned to the given countries*

`--input topology.txt[.gz] or .gml --countries UY,AR,BR --output base`
`--date YYYYMMDD (historical NRO snapshot) --justify-output`

Always produces .txt + .gml (nodes carry country attr) · ASNs not in NRO file are excluded · ISO 3166-1 alpha-2 codes

annotate-topology.py *Add country + RIR columns to every edge*

`--input topology.txt[.gz] --output topology-annotated.txt --date YYYYMMDD --justify-output`

Output: 6057 3359 uy lacnic us arin · Unknown/private ASNs → unknown unknown
RIR names: arin · ripencc · apnic · lacnic · afrinic · Country codes lowercase ISO 3166-1 alpha-2

NRO delegated stats (used by geo-filter, annotate-topology, build-topology --filter-registered)
Cached next to input: `nro-delegated-stats` (latest, re-fetched if >24 h old) · `nro-delegated-stats-YYYYMMDD` (historical, permanent)

AS path rules (extract-paths & download-ris-day): prepending collapsed (A A B → A B) · AS sets truncate path · edges stored as `(min, max)` integer pair — not lexicographic